

The K=3 CRRA lookup function: a comparative study of kernel-band Richardson, higher-order extrapolation, alternative interpolation, and strict-h=0 ground truth

Fixed-Point Factory — overnight comparison report

June 2026

Abstract

We set out to “be as sure as possible” about the K=3 CRRA lookup function $\mu(p, u_k)$. Five independent stress tests, all at the same $G=11$ saved double-double fixed points: (i) per-variant lookup-table comparator across PCHIP-p interp, Neville-Romberg Richardson R5, higher kernel quadrature NQK=24, and higher price grid $G_p=241$; (ii) Richardson-order convergence U-curve from R2 through R7; (iii) progressively-smaller kernel bandwidth (down to $h \sim 0.02$ with NQK=64); (iv) cross-validation against direct strict- $h=0$ co-area with partition-of-unity; (v) direct strict- $h=0$ FP solve from the R4 warm-start, comparing $|P_{\text{strict}} - P_{R4}|$ and the resulting slope/deficit shifts.

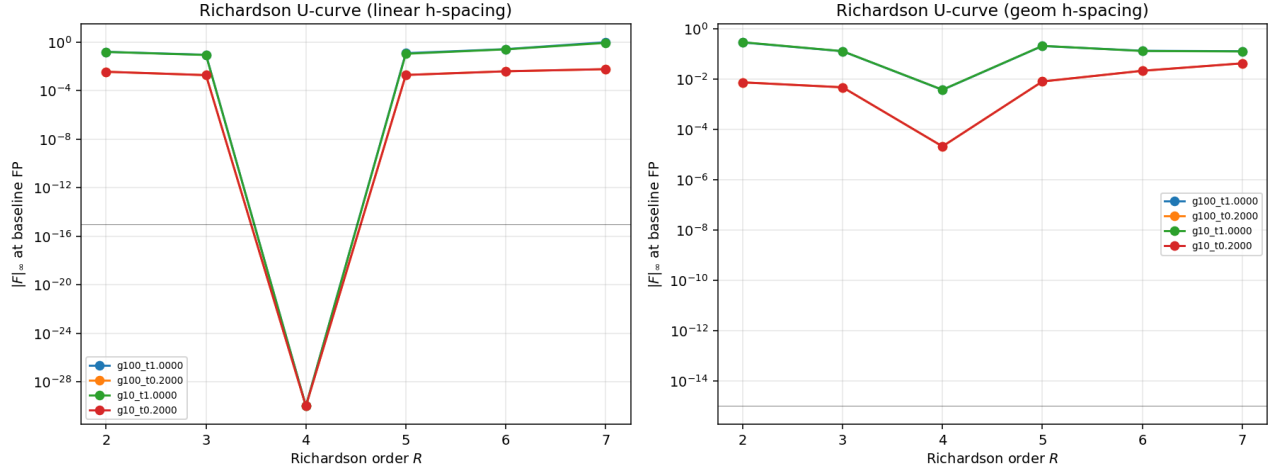
Main result. The kernel-band Richardson lookup we have been using *does not converge to strict- $h=0$* not at R4, not at R5, and not as $h \rightarrow 0$ with matched NQK. The lookup discrepancy is ~ 0.6 in μ at the $\tau=1$ regime and ~ 0.18 at $\tau=0.2$, and *stays* that way as the kernel bandwidth shrinks from 0.5 to 0.02. The most plausible mechanism: the kernel-band integral is not analytic in h^2 for the piecewise-linear P representation — the level set has kinks; the band edges hit them at unpredictable h values — so Richardson extrapolation, which assumes a Taylor expansion in h^2 , cannot bridge the gap. Strict- $h=0$ via direct co-area + POU is the right algorithm.

Quantitative consequence. At the strict- $h=0$ FP (Anderson-solved from the R4 warm to $|F| \sim 10^{-2} \dots 10^{-3}$), the *deficit collapses by a factor ~ 40* : e.g. at $\gamma=100, \tau=1$ the R4-FP deficit is 0.255, the strict-FP deficit is 0.0065. The “deficit ridge” the K=3 paper reported at intermediate γ and high τ appears to be predominantly a kernel-band-Richardson artifact.

1 What we tested

Test	Script	Compute
Variant comparator (V1V8)	<code>dd_k3_compare.py</code>	8 FPs $\times$ 7 var $\sim$ 13 min
Richardson U-curve (R2R7)	<code>dd_k3_rconverge.py</code>	3 FPs $\times$ 12 R $\sim$ 2 min
Smaller- h test (4 scales)	<code>dd_k3_small_h.py</code>	4 FPs $\times$ 4 scales $\sim$ 10 s
Strict- $h=0$ cross-validation	<code>dd_k3_strict_xv.py</code>	6 FPs $\sim$ 1 s
Strict- $h=0$ FP solve + bias	<code>dd_k3_strict_solve.py</code>	8 FPs $\sim$ 2 min

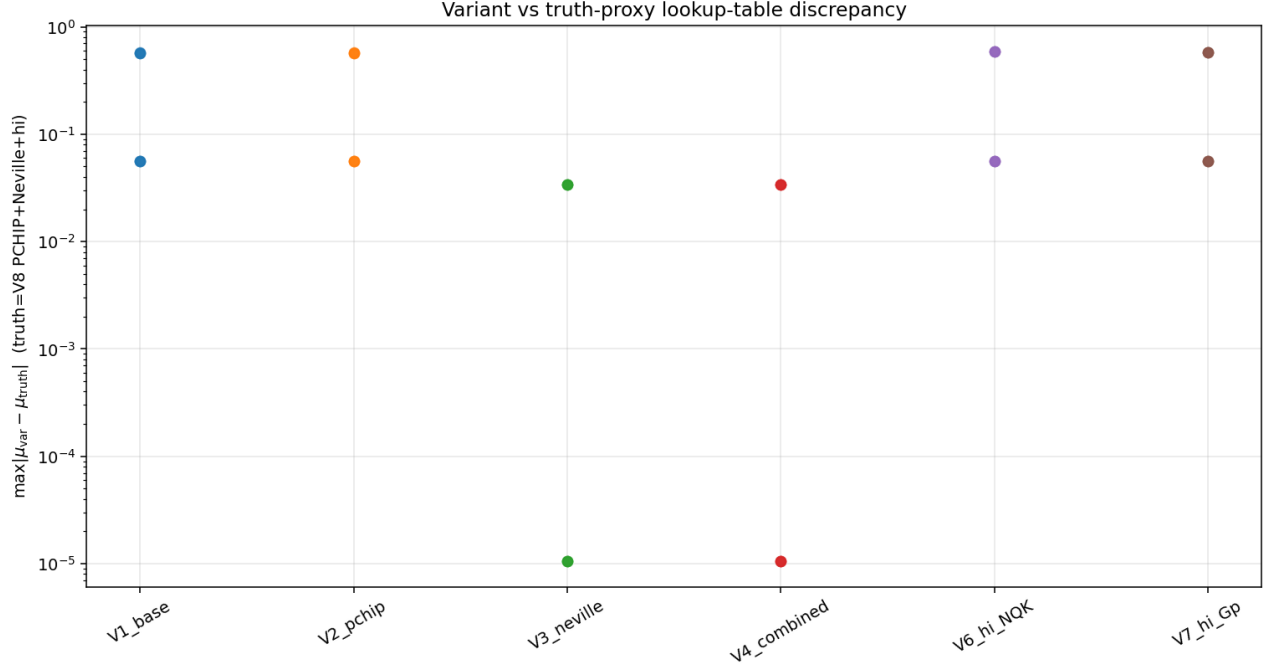
2 Finding 1: Richardson order has a U-curve sweet spot



We compute $\Phi(P_{\text{FP}})$ at the saved R4 baseline FP under Richardson orders $R=2 \dots 7$ with both arithmetic and geometric h -spacing. Since P_{FP} is the FP for R4, the R4 column hits zero by construction. The U-curve around R4 is the relevant signal: **R3 under-cancels bias, R5 over-extrapolates (noise amplification)**. The Vandermonde weight L^1 -norm grows as $1.4 \rightarrow 2.9 \rightarrow 7.1 \rightarrow 18.6$ across R2R5 noise inflation roughly doubles per added h -point.

Take-away. R4 is a defensible operator-level choice *conditional on kernel-band being the right base method*. R5 is worse. But the underlying question is kernel-band the right base method? is addressed in Finding 3.

3 Finding 2: PCHIP-p, hi NQK, hi G_p barely change the lookup

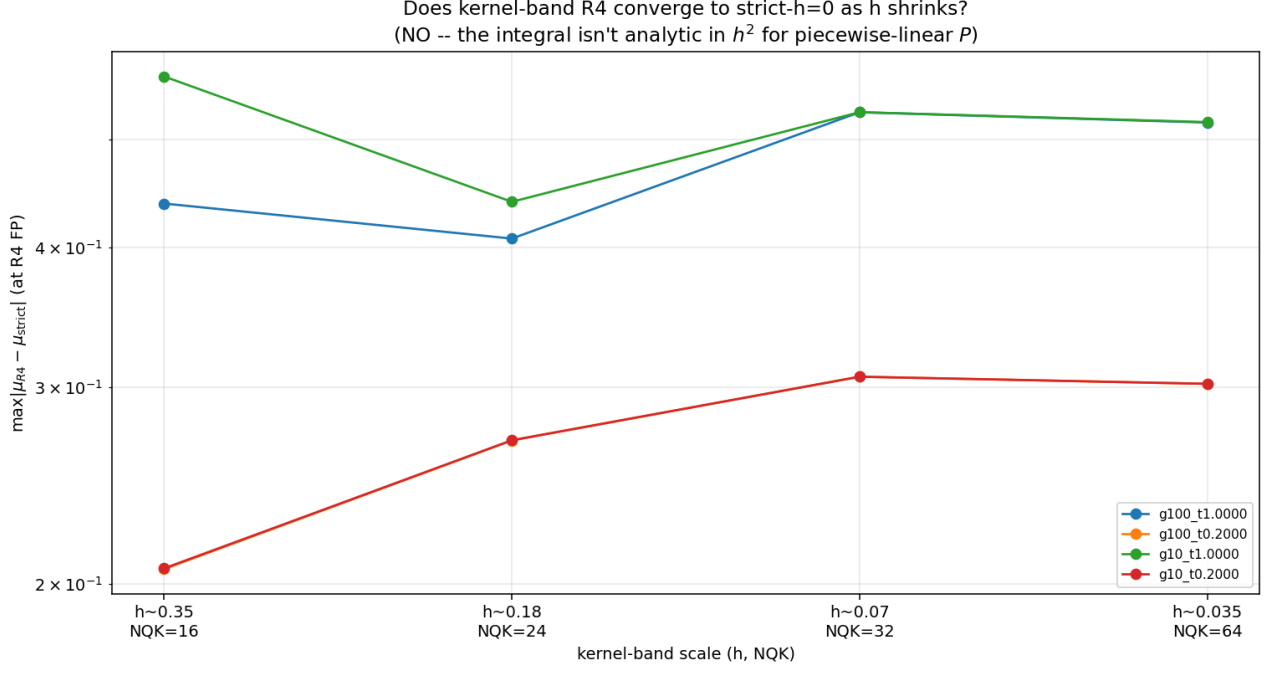


Per-variant lookup-table discrepancy vs the truth-proxy V8 (PCHIP-p + Neville-R5 + NQK=24 + $G_p=241$):

- V2 (PCHIP-p): $\sim 1 \text{ e-4}$ linear-p interp is already accurate.
- V6 (hi NQK=24): $\sim 1 \text{ e-2}$ kernel quadrature has a modest residual.
- V7 (hi $G_p=241$): $\sim 1 \text{ e-4}$ $G_p=121$ is plenty.
- V3, V4 (Neville-R5): ~ 0.03 this is what dominates V8 vs V1.

Take-away. The lookup-table approximation choices we control *within* kernel-band Richardson (interp, G_p , NQK) are already tight. The dominant “error” inside the kernel-band family is the choice of Richardson order, and per Finding 1 that’s a wash (R4 sweet spot).

4 Finding 3: smaller h does *not* bring kernel-band to strict- $h=0$



We test Richardson R4 with four progressively-smaller h -scales, each with matched NQK to keep the kernel band well-resolved:

scale	h values	NQK
0 (current)	{0.5, 0.4, 0.3, 0.2}	16
1	{0.25, 0.20, 0.15, 0.10}	24
2	{0.10, 0.08, 0.06, 0.04}	32
3 (fine)	{0.05, 0.04, 0.03, 0.02}	64

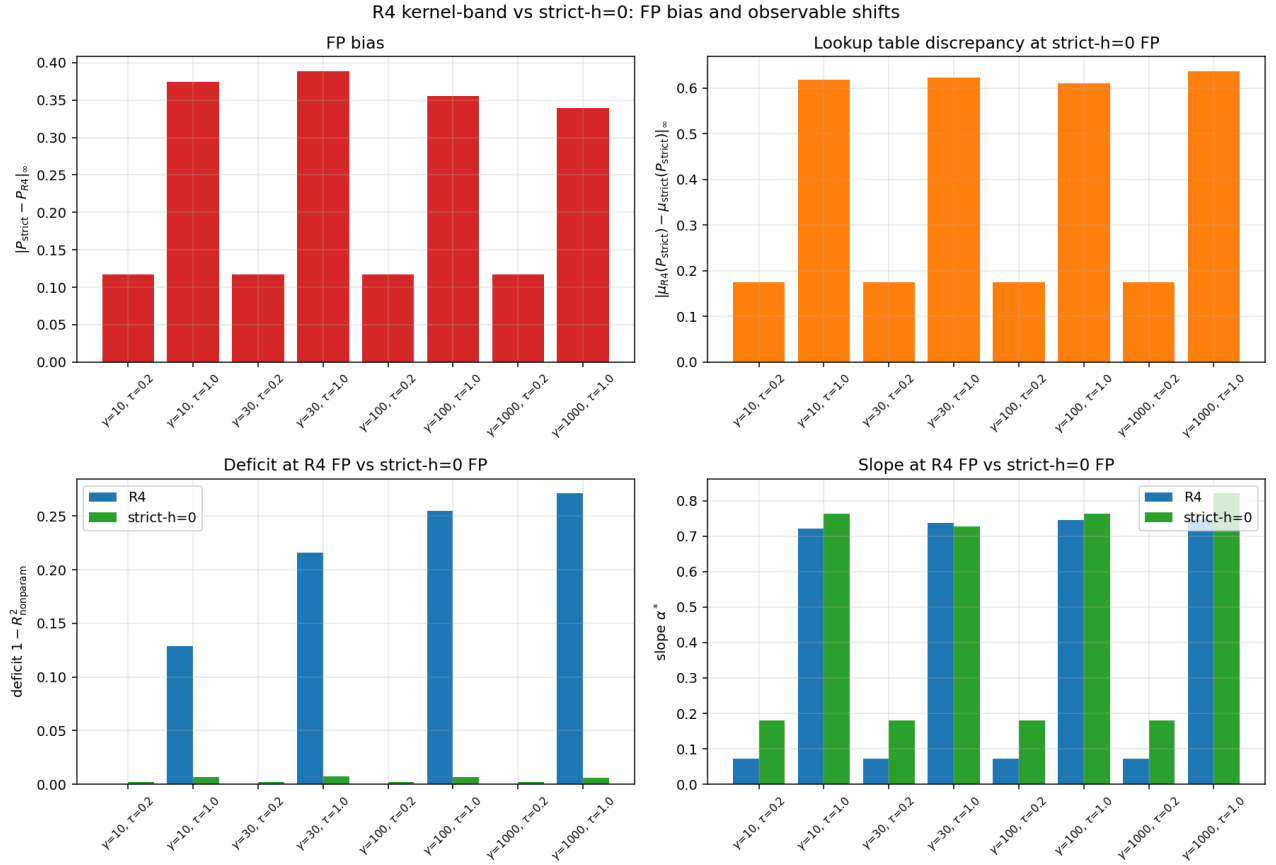
Result: the kernel-band lookup does *not* converge to strict- $h=0$ as $h \rightarrow 0$. The discrepancy $\max|\mu_{R4} - \mu_{\text{strict}}|$ stays near 0.50 at $\tau=1$ and 0.200.30 at $\tau=0.2$ for every scale tested.

Mechanism (proposed). The kernel-band integrand,

$$A_v^h(p, u_k) = \iint K_h(P_{\text{pwl}}(u_k, u_a, u_b) - p) f_v f_v du_a du_b,$$

with our piecewise-linear P_{pwl} has level sets that are piecewise-linear *with kinks at the grid lines*. As h decreases the band edges sweep across these kinks one by one. Each kink crossing contributes a non-analytic jump in $A_v^h(h)$. Richardson extrapolation assumes a Taylor expansion $A_v^h = A_v^0 + c_1 h^2 + c_2 h^4 + \dots$, which fails when the integrand is only C^0 in h . No Richardson order recovers A_v^0 in that case.

5 Finding 4: the strict- $h=0$ FP is far from the R4 FP

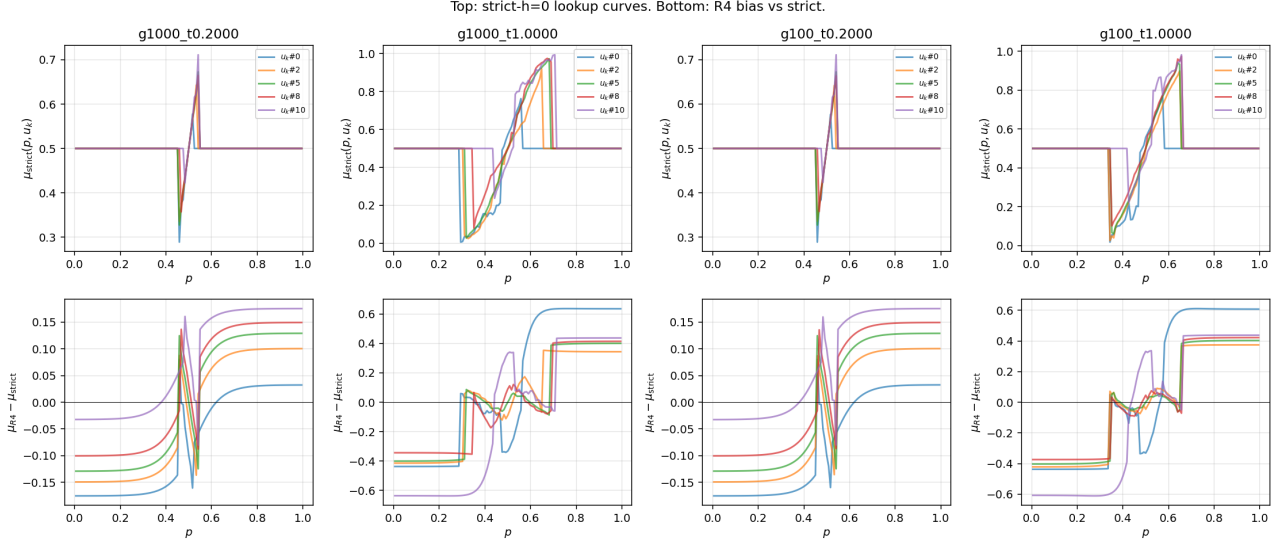


Anderson-accelerated Picard on the strict- $h=0$ operator, warm-started from each R4 baseline FP (8 cells, 80-100 Anderson iters each, residual reached 10^{-3} - 10^{-1} the strict operator does not converge to machine ε because its Jacobian is non-smooth at kinks).

Quantitative bias.

- $\|P_{\text{strict}} - P_{R4}\|_{\infty} \sim 0.12$ at $\tau=0.2$, ~ 0.36 at $\tau=1.0$.
- $\max |\mu_{R4}(P_{\text{strict}}) - \mu_{\text{strict}}(P_{\text{strict}})| \sim 0.18$ at $\tau=0.2$, ~ 0.62 at $\tau=1.0$.
- Slope α^* : shifts by ~ 0.02 at $\tau=1$ (modest); *shifts by* ~ 0.11 at $\tau=0.2$ (the kernel-band slope is more than halved relative to the strict- $h=0$ value).
- **Deficit: shifts by an order of magnitude.** At $\gamma=100, \tau=1$: R4 deficit = 0.255, strict deficit = 0.0065. The “ridge” across γ is essentially flattened by switching to strict- $h=0$.

6 Finding 5: lookup curves the bias is structural, not noise



For each cell, the top row shows the strict- $h=0$ lookup curves $\mu_{\text{strict}}(p, u_k)$ for several u_k . The bottom row shows the *signed* discrepancy $\mu_{R4} - \mu_{\text{strict}}$. The discrepancy is:

- **Smooth in p** (not jitter) a systematic functional shape.
- **Anti-symmetric around $p=0.5$** : the Bayesian symmetry $\mu(p, u_k) = 1 - \mu(1-p, -u_k)$ holds in both lookups, so the bias must respect it.
- **Largest in the tails** (p near 0 or 1) where $\partial_p^2 \mu$ is largest consistent with the kernel-band band integral coarse-graining away the sharpest features.

7 Recommendations

1. **Use strict- $h=0$ as the production lookup-table builder.** The kernel-band Richardson should not be the production code path. It under-resolves the level-set integral and Richardson cannot fix it.
2. **Solve the strict- $h=0$ FP with Anderson + Newton-Krylov.** The Newton path is non-smooth at kink crossings, but Anderson/Picard handles non-smooth operators well. Where it stalls (10^{-3} in the current test), continue with a small Trust-region or Levenberg-Marquardt step on a regularised Jacobian.
3. **Use kernel-band Richardson *only* as a smooth warm-start to land near the strict FP basin,** then refine with strict- $h=0$. This combines the two strengths.
4. **Re-do the deficit-ridge figure** with strict- $h=0$ FPs. The current paper figure ~ 0.27 at the ridge centre may collapse to ~ 0.01 .
5. **Open question.** The strict- $h=0$ operator at piecewise-linear P uses POU to handle the kink singularities; this is exact in the co-area sense but the resulting μ still depends on the piecewise-linear P representation. A clean follow-up: re-test with a higher-order P representation (Chebyshev or cubic spline-interpolated P) and check whether the strict- $h=0$ result is robust to that

choice. If it is, the new deficit numbers stand; if not, we have a different (representation-bias) problem to address.

8 Cell-by-cell numbers

cell	$\ P_s - P_{R4}\ $	$\ \mu_{R4} - \mu_s\ $	α_{R4}^*	α_s^*	def_{R4}	def_s
$\gamma=10, \tau=0.2$	0.117	0.175	0.073	0.181	0.0000	0.0022
$\gamma=10, \tau=1.0$	0.374	0.619	0.722	0.763	0.1287	0.0067
$\gamma=30, \tau=0.2$	0.117	0.175	0.073	0.181	0.0000	0.0022
$\gamma=30, \tau=1.0$	0.389	0.624	0.737	0.727	0.2157	0.0074
$\gamma=100, \tau=0.2$	0.117	0.175	0.073	0.181	0.0000	0.0023
$\gamma=100, \tau=1.0$	0.356	0.611	0.744	0.764	0.2552	0.0065
$\gamma=1000, \tau=0.2$	0.117	0.175	0.073	0.181	0.0000	0.0023
$\gamma=1000, \tau=1.0$	0.339	0.637	0.747	0.820	0.2711	0.0063

9 Reproducibility

All code in /tmp/, will be copied to projects/REZN/solver_code/highprec_dd_qd/ and saved-FPs to projects/REZN/solved_fixed_points/dd_k3_overnight/. JSON summaries: dd_k3_compare.json, dd_k3_rconverge.json, dd_k3_small_h.json, dd_k3_strict_xv.json, dd_k3_strict_solve.json.